

Design Concepts Transcript

Now, we will be looking at design concepts. Design concepts are here to help you to do build more functionality into your visualisations. Let me take a pause here and explain.

The design concepts that we will be discussing today for building effective visualisations borrows heavily from the product design philosophy. The products that are designed for our day to day use, the most useful and successful products have a wider acceptance, that is because they adopt the principle form follows the function. To design a particular product, the product managers or the designers have always ask the question what is the function this particular product is expected to deliver.

By focussing on the function that is delivering, a traditionally best products have been designed. The visualisation function is to effectively communicate the information with the audience. We also adopt the same philosophy for developing the effective visuals.

We have to ask what is the function of our visualisation. Again keeping audience in the centre, we try to design the visualisations and in this process we follow three steps. First is create affordances which are best suited to convey the information.

Then we think about the accessibility of the information that we are presenting to a wider range of audience and we also take care of the aesthetics, so that the visualisation is pleasing to wide range of audience and they are ready to receive the information that it is providing. All these put together would lead to higher acceptance of the visual from the audience. Therefore, creating a smooth transformation of information that you intend to present in the visual and perceived by the audience.

This is again following the product design philosophy, it simply says the best products are the ones which give out the functions which they are intended to and then design the form that efficiently delivers that function. Let us deep dive into each of these concepts. First we will talk about affordances.

What are affordances? Affordances are artefacts that help enhance your visual. For example, you can use simple text highlighting with bold or colours to deliver impact on your visuals. At the same time, you can declutter and simplify the process of the visualisation, so that you can clearly present your message.

Let me walk you through this with an example. Consider the first visualisation that I am highlighting. So, this is new marriage by education.

So, you have seen all the education rate is I mean the marriage rate is 41.4 percent compared to 2008 versus 2012 in and it is 36.7 percent. The way to read the bar graph though it is intuitive is very difficult because you have to first spend energy on placing the axis and the colour legend which is at the top and have to drive the intuition saying that marriage rates have fallen as with the increase in the education. This is the conclusion that the visual is trying to showcase.

However, it is very difficult or at least it is taxing to drive this particular information from the graphic. The same graphic can be made more affordance using affordances by simplifying the axis, you are presenting the years in the axis now and taking the degree category, the educational category and simply posting two points. Instead of showing the transformation between, you are just showing the start point and end point and highlighting them with ready variables, ready labels, so that you can show that with earlier education the marriage rates are falling and in general the marriage rate has fallen and you can write the text somewhere here in between to create even more impact.

This is what affordances all about. For example, you have a complex visual, you can simplify that and become it and make it more affordable. You can transform a complex visual by simplifying its presentation, by transforming the axis and using text and labelling to deliver the information that you want to convey in most effective format and reducing the processing time for the audience.

Therefore, they spend less mental energy and grab more information out of the visual. The second concept within affordances is creating visual hierarchies in the information. So for example, there is a scatter plot which shows the satisfaction and of the car users and the number of issues they have faced in the car.

It is a simple 2 by 2 scatter plot where you have number of issues on one axis and the satisfaction level on the other axis and you can create a layer of sophistication on this hierarchy by saying that we can divide the number of issues at an average level or say at a particular level and say ones which are above this level are saying to have higher number of issues versus the ones below this have lesser number of issues. Simultaneously, we can also create a category on the satisfaction axis to show that we have highly satisfied customers with feedback ratings of say 80% or above 75% or above and the ones with less than 70% feedback rating are unsatisfied customers or customers with low satisfaction. This is enabling us to segment the data into 4 buckets that is highly satisfied customers with high issues, highly satisfied customers with low issues, low satisfied customers with high issues, low satisfied customers with low issues.

This helps the business managers to focus on particular segment. We need 3 segments and they constitute 50% of your customer base and the rest are highlighted with grey in colour. These customers are 50% of your customer base which are highlighted prominently and other segments have been used with lighter colour showing that they are not that important or lesser in your customer base.

So, this is more intuitive approach and aesthetically pleasing not taxing on your eyes. So, it is not just sufficient to do a good visualisation, but one has to take care whether it is aesthetically pleasing for presenting. So, these aspects like using the right combination of colours and highlighting the appropriate segments within the visual.

So, that there is an aesthetic flavour to the visualisation has to be learned with practise



and you have to keep in mind once we present things better it is received better and the communication gets more effectively. So, finally, we will talk about a concept called acceptance. Acceptance is when you present the visual how is it being received by the audience.

Acceptance is very important because you have to finally, able to convey your information to the audience and they should receive the information in the intended form. They might receive something else or present questions that are not intended from that particular visual. So, it is very important to understand what needs to be built to get that acceptance.

While designing the visual or interacting with the audience you have to keep in mind it is not easy to accept change. People always resist change if you are presenting data in a new format or providing a new insight it is first the duty of the analyst or the person who is presenting the data or the visualisation to create acceptance in the audience. How can you do this? Can explain the rationale and you can show the comparison and how the current visual is better than the earlier visualisation or the current insight is better than the earlier insight and maybe take support from couple of audience who have understood your point and make them your vocal member try to gain support and also try to take constructive feedback and iterate the process and engage with the audience to get the broader acceptance.

If you use proper affordances ensure accessibility and make it aesthetically pleasing getting acceptance would be far more easier if these three elements are not present. There is situation where your visualisation is not aesthetically pleasing or it misses ah it is very tough to access for a broader range of audience or it is very ah cluttered and it is not easily ah readable to the audience getting acceptance would become far more harder. So, once you take care of the design concepts acceptance should normally follow.

With this you can do better visualisations keep this design concepts in your mind as you go forward and design your visuals.

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